Toward the Császár Polyhedron via Cell-Complex Enumeration

Step 1: Notation and the First Conditions

Notation

Let $P_i = (x_i, y_i, z_i) \in \mathbb{R}^3$ be the coordinate vector of the symbolic point i. Define the *bracket*

$$[abcd] \;:=\; \det\begin{pmatrix} 1 & x_a & y_a & z_a \\ 1 & x_b & y_b & z_b \\ 1 & x_c & y_c & z_c \\ 1 & x_d & y_d & z_d \end{pmatrix} \;=\; \det\bigl(P_b - P_a,\; P_c - P_a,\; P_d - P_a\bigr),$$

which is 6 times the signed volume of the tetrahedron $abcd$ and is alternating in a, b, c, d. With this,

$$abc^{\pm} \;:=\; \bigl\{\, p \in \mathbb{R}^3 \;:\; \pm[a, b, c, p] > 0 \,\bigr\}.$$

Condition 1: $3 \in 012^{+}$

This is just $[0, 1, 2, 3] > 0$. Three equivalent forms:

(a) Determinant.

$$\det\begin{pmatrix} 1 & x_0 & y_0 & z_0 \\ 1 & x_1 & y_1 & z_1 \\ 1 & x_2 & y_2 & z_2 \\ 1 & x_3 & y_3 & z_3 \end{pmatrix} > 0 \quad\Longleftrightarrow\quad \det\begin{pmatrix} x_1 - x_0 & y_1 - y_0 & z_1 - z_0 \\ x_2 - x_0 & y_2 - y_0 & z_2 - z_0 \\ x_3 - x_0 & y_3 - y_0 & z_3 - z_0 \end{pmatrix} > 0$$

(b) Scalar triple product (cross-product antisymmetry).

$$(P_1 - P_0) \cdot \bigl((P_2 - P_0) \times (P_3 - P_0)\bigr) \;>\; 0.$$

This form privileges P_0 as a basepoint, hiding the S_4-antisymmetry on the four points. *Fully unparenthesized.* Distribute the differences (using $a{\times}a = 0$ and the cyclic identity $a{\cdot}(b{\times}c) = c{\cdot}(a{\times}b)$); the bracket reorganizes into the alternating sum over 3-element subsets of $\{0, 1, 2, 3\}$:

$$[0, 1, 2, 3] \;=\; P_1{\cdot}(P_2{\times}P_3) \;-\; P_0{\cdot}(P_2{\times}P_3) \;+\; P_0{\cdot}(P_1{\times}P_3) \;-\; P_0{\cdot}(P_1{\times}P_2).$$

Equivalently, with $T(i, j, k) := P_i \cdot (P_j \times P_k)$ (cyclically symmetric in i, j, k) and $\widehat{d}$ denoting the elements of $\{0, 1, 2, 3\} \setminus \{d\}$ in increasing order,

$$[0, 1, 2, 3] \;=\; \sum_{d=0}^{3} (-1)^d\, T\bigl(\widehat{d}\bigr).$$

The S_4-antisymmetry is now manifest — any swap of two points flips the bracket's sign.

(c) Polynomial in coordinates. With $\Delta_i := P_i - P_0$ for $i = 1, 2, 3$:

$$\Delta_{1,x}(\Delta_{2,y}\Delta_{3,z} - \Delta_{2,z}\Delta_{3,y}) + \Delta_{1,y}(\Delta_{2,z}\Delta_{3,x} - \Delta_{2,x}\Delta_{3,z}) + \Delta_{1,z}(\Delta_{2,x}\Delta_{3,y} - \Delta_{2,y}\Delta_{3,x}) \;>\; 0.$$

Expanding the Δ's as differences of original coordinates:

$$\begin{aligned} &(x_1{-}x_0)\big[(y_2{-}y_0)(z_3{-}z_0) - (z_2{-}z_0)(y_3{-}y_0)\big] \\ +\;&(y_1{-}y_0)\big[(z_2{-}z_0)(x_3{-}x_0) - (x_2{-}x_0)(z_3{-}z_0)\big] \\ +\;&(z_1{-}z_0)\big[(x_2{-}x_0)(y_3{-}y_0) - (y_2{-}y_0)(x_3{-}x_0)\big] \;>\; 0. \end{aligned}$$

Again the S_4-antisymmetry is hidden inside the differences. *Fully unparenthesized.* Distributing all subtractions yields the 24-term Levi-Civita expansion:

$$[0,1,2,3] \;=\; \sum_{(d,a,b,c)\in S_4} \operatorname{sgn}(d,a,b,c)\, x_a\, y_b\, z_c,$$

where (d, a, b, c) ranges over permutations of $(0, 1, 2, 3)$ and d is the row contributing the constant 1 from the bordered determinant (so d enters only through the sign). Grouped by missing index d:

$$\begin{aligned} d{=}0: &\quad +\,x_1y_2z_3 - x_1y_3z_2 - x_2y_1z_3 + x_2y_3z_1 + x_3y_1z_2 - x_3y_2z_1 \\ d{=}1: &\quad -\,x_0y_2z_3 + x_0y_3z_2 + x_2y_0z_3 - x_2y_3z_0 - x_3y_0z_2 + x_3y_2z_0 \\ d{=}2: &\quad +\,x_0y_1z_3 - x_0y_3z_1 - x_1y_0z_3 + x_1y_3z_0 + x_3y_0z_1 - x_3y_1z_0 \\ d{=}3: &\quad -\,x_0y_1z_2 + x_0y_2z_1 + x_1y_0z_2 - x_1y_2z_0 - x_2y_0z_1 + x_2y_1z_0 \end{aligned}$$

Every monomial uses each index at most once, and any swap of two indices flips the signs of all 24 terms in lock-step — the four-fold antisymmetry on $\{0, 1, 2, 3\}$ is entirely visible.

Condition 2: $012^+ \cap 013^+ \neq \emptyset$

Existence of p with $[0, 1, 2, p] > 0$ *and* $[0, 1, 3, p] > 0$.

(a) Determinant.

$$\exists\, p \in \mathbb{R}^3: \quad \det\begin{pmatrix} 1 & x_0 & y_0 & z_0 \\ 1 & x_1 & y_1 & z_1 \\ 1 & x_2 & y_2 & z_2 \\ 1 & p_x & p_y & p_z \end{pmatrix} > 0 \;\wedge\; \det\begin{pmatrix} 1 & x_0 & y_0 & z_0 \\ 1 & x_1 & y_1 & z_1 \\ 1 & x_3 & y_3 & z_3 \\ 1 & p_x & p_y & p_z \end{pmatrix} > 0$$

(b) Scalar triple product.

$$\exists\, p \in \mathbb{R}^3: \quad (P_1{-}P_0)\cdot\big((P_2{-}P_0)\times(p{-}P_0)\big) > 0 \;\wedge\; (P_1{-}P_0)\cdot\big((P_3{-}P_0)\times(p{-}P_0)\big) > 0.$$

Fully unparenthesized. Each bracket cofactor-expands as in Condition 1(b), with p playing the role of the fourth point:

$$\begin{aligned} [0,1,2,p] &= P_1{\cdot}(P_2{\times}p) \;-\; P_0{\cdot}(P_2{\times}p) \;+\; P_0{\cdot}(P_1{\times}p) \;-\; P_0{\cdot}(P_1{\times}P_2), \\ [0,1,3,p] &= P_1{\cdot}(P_3{\times}p) \;-\; P_0{\cdot}(P_3{\times}p) \;+\; P_0{\cdot}(P_1{\times}p) \;-\; P_0{\cdot}(P_1{\times}P_3). \end{aligned}$$

The shared term $P_0{\cdot}(P_1{\times}p)$ reflects that both brackets share the edge $\overline{P_0P_1}$. Each is S_4-antisymmetric in its own four points ($\{0, 1, 2, p\}$ resp. $\{0, 1, 3, p\}$) and *affine* in p.

(c) Polynomial in coordinates. Substitute p for P_3 in the first inequality, and p for P_2 (keeping P_3 in the third slot) in the second. Writing $p = (x_p, y_p, z_p)$:

$$\exists\,(x_p, y_p, z_p) \in \mathbb{R}^3 :$$

$$\begin{aligned}
&(x_1{-}x_0)\big[(y_2{-}y_0)(z_p{-}z_0) - (z_2{-}z_0)(y_p{-}y_0)\big] \\
&+ (y_1{-}y_0)\big[(z_2{-}z_0)(x_p{-}x_0) - (x_2{-}x_0)(z_p{-}z_0)\big] \\
&+ (z_1{-}z_0)\big[(x_2{-}x_0)(y_p{-}y_0) - (y_2{-}y_0)(x_p{-}x_0)\big] \;>\; 0
\end{aligned}$$

AND

$$\begin{aligned}
&(x_1{-}x_0)\big[(y_3{-}y_0)(z_p{-}z_0) - (z_3{-}z_0)(y_p{-}y_0)\big] \\
&+ (y_1{-}y_0)\big[(z_3{-}z_0)(x_p{-}x_0) - (x_3{-}x_0)(z_p{-}z_0)\big] \\
&+ (z_1{-}z_0)\big[(x_3{-}x_0)(y_p{-}y_0) - (y_3{-}y_0)(x_p{-}x_0)\big] \;>\; 0.
\end{aligned}$$

Fully unparenthesized. Each bracket admits the 24-term Levi-Civita expansion of Condition 1(c). For $[0, 1, 2, p]$, treating p as the fourth point and grouping by missing index:

$$\begin{aligned}
d{=}0: &\quad + x_1y_2z_p - x_1y_pz_2 - x_2y_1z_p + x_2y_pz_1 + x_py_1z_2 - x_py_2z_1 \\
d{=}1: &\quad - x_0y_2z_p + x_0y_pz_2 + x_2y_0z_p - x_2y_pz_0 - x_py_0z_2 + x_py_2z_0 \\
d{=}2: &\quad + x_0y_1z_p - x_0y_pz_1 - x_1y_0z_p + x_1y_pz_0 + x_py_0z_1 - x_py_1z_0 \\
d{=}p: &\quad - x_0y_1z_2 + x_0y_2z_1 + x_1y_0z_2 - x_1y_2z_0 - x_2y_0z_1 + x_2y_1z_0.
\end{aligned}$$

$[0, 1, 3, p]$ is obtained by replacing every subscript 2 with 3 above. Two structural observations are now visible:

- Rows $d = 0, 1, 2$ contain all the p-dependence (linear in p); row $d = p$ is the constant $-\det(P_0, P_1, P_2)$ (resp. $-\det(P_0, P_1, P_3)$). So each bracket is affine in p, and the realizability question is whether two open halfspaces in p-space meet.
- The S_4-antisymmetry on the bracket's four points $\{0, 1, 2, p\}$ (or $\{0, 1, 3, p\}$) is manifest: every swap of two indices flips the signs of all 24 monomials simultaneously.

Realizability: $012^+ \cap 013^+$ exists, $012^- \cap 013^+ \cap 123^+ \cap 203^+$ does not

Notation conversion. The bracket $[a, b, c, d]$ is alternating, so $[2, 0, 3, p] = -[0, 2, 3, p]$, hence

$$203^+ \;=\; \{p : [2, 0, 3, p] > 0\} \;=\; \{p : [0, 2, 3, p] < 0\} \;=\; 023^-.$$

The cell in question is therefore $012^- \cap 013^+ \cap 023^- \cap 123^+$ in canonical (sorted-index) notation — the sign vector $(-, +, -, +)$ on the four faces $(012, 013, 023, 123)$ of tetrahedron 0123.

Key identity. For any $p \in \mathbb{R}^3$,

$$[0, 1, 2, p] \;-\; [0, 1, 3, p] \;+\; [0, 2, 3, p] \;-\; [1, 2, 3, p] \;=\; [0, 1, 2, 3]. \qquad (\star)$$

Proof. Since P_0, P_1, P_2, P_3 are affinely independent (because $[0, 1, 2, 3] \neq 0$), every $p \in \mathbb{R}^3$ has unique barycentric coordinates

$$p \;=\; \alpha_0P_0 + \alpha_1P_1 + \alpha_2P_2 + \alpha_3P_3, \qquad \alpha_0 + \alpha_1 + \alpha_2 + \alpha_3 \;=\; 1.$$

The lifted equality $(1,p) = \sum_i \alpha_i\,(1,P_i)$ holds in $\mathbb{R}^4$, so multilinearity of the bordered 4×4 determinant in row p gives, for distinct $a,b,c \in \{0,1,2,3\}$ with missing index d,

$$[a,b,c,p] \;=\; \sum_{i=0}^{3} \alpha_i\,[a,b,c,i] \;=\; \alpha_d\,[a,b,c,d],$$

since $[a,b,c,a] = [a,b,c,b] = [a,b,c,c] = 0$ (repeated row). Using $[0,1,3,2] = -[0,1,2,3]$, $[0,2,3,1] = +[0,1,2,3]$, $[1,2,3,0] = -[0,1,2,3]$ (reorder the missing index back into canonical position), and dividing by $[0,1,2,3]$:

$$\alpha_3 \;=\; \frac{[0,1,2,p]}{[0,1,2,3]}, \quad \alpha_2 \;=\; \frac{-[0,1,3,p]}{[0,1,2,3]}, \quad \alpha_1 \;=\; \frac{[0,2,3,p]}{[0,1,2,3]}, \quad \alpha_0 \;=\; \frac{-[1,2,3,p]}{[0,1,2,3]}.$$

Then $\alpha_0+\alpha_1+\alpha_2+\alpha_3 = 1$, multiplied through by $[0,1,2,3]$, is exactly $(\star)$. $\square$

Why $012^+ \cap 013^+$ is non-empty. Take the explicit witness

$$p^* \;:=\; 2P_3 - P_2 \qquad \text{(the reflection of } P_2 \text{ across } P_3\text{)}.$$

Its coefficients $(2,-1)$ sum to 1, so any affine function of p evaluated at p^* equals 2 times its value at P_3 minus its value at P_2. With $[0,1,2,P_2] = 0$ and $[0,1,2,P_3] = [0,1,2,3]$:

$$[0,1,2,\,p^*] \;=\; 2\,[0,1,2,3] \;-\; 0 \;=\; 2\,[0,1,2,3] \;>\; 0.$$

With $[0,1,3,P_2] = [0,1,3,2] = -[0,1,2,3]$ and $[0,1,3,P_3] = 0$:

$$[0,1,3,\,p^*] \;=\; 2\cdot 0 \;-\; (-[0,1,2,3]) \;=\; [0,1,2,3] \;>\; 0.$$

Both brackets are positive, so $p^* \in 012^+ \cap 013^+$. (Geometrically: planes 012 and 013 are distinct and share the line $\overline{P_0P_1}$, so they cut $\mathbb{R}^3$ into four open wedges, all non-empty; $012^+ \cap 013^+$ is one of them.)

Why $012^- \cap 013^+ \cap 023^- \cap 123^+$ is empty. Suppose for contradiction such a p exists. The four memberships translate to

$$[0,1,2,p] < 0, \quad [0,1,3,p] > 0, \quad [0,2,3,p] < 0, \quad [1,2,3,p] > 0.$$

Hence each summand on the left side of $(\star)$, with its prescribed sign, is *strictly negative*:

$$\underbrace{[0,1,2,p]}_{<0} \;\underbrace{-\,[0,1,3,p]}_{<0} \;\underbrace{+\,[0,2,3,p]}_{<0} \;\underbrace{-\,[1,2,3,p]}_{<0} \;<\; 0.$$

But by $(\star)$ this sum equals $[0,1,2,3]$, which is >0 by Condition 1. Contradiction. Hence $012^- \cap 013^+ \cap 023^- \cap 123^+ = \emptyset$. $\square$

Geometric meaning. In barycentric coordinates the four sign conditions read

$$\alpha_0 < 0, \quad \alpha_1 < 0, \quad \alpha_2 < 0, \quad \alpha_3 < 0 \qquad (\text{“}p \text{ on the wrong side of every face”}),$$

which is impossible because $\alpha_0+\alpha_1+\alpha_2+\alpha_3 = 1 > 0$. The four face-planes of 0123 in general position cut $\mathbb{R}^3$ into 15 open cells (the count $\binom{4}{0}+\binom{4}{1}+\binom{4}{2}+\binom{4}{3}$), and the missing 16th sign vector is precisely $(012^-, 013^+, 023^-, 123^+)$. The asymmetry between the two cells of the question is now visible: $012^+ \cap 013^+$ slices $\mathbb{R}^3$ along only *two* planes (a wedge — always non-empty), while the missing 16th cell asks for a sign-coherent intersection across *all four* faces of a closed tetrahedron, which the affine relation $\sum \alpha_i = 1$ rules out.

Observations

- **Only signs matter.** The cell complex depends only on the *sign vector* in $\{+,-\}^{\binom{n}{3}}$. Coordinates are useful only as a realizability oracle: if we want a "without coordinates" enumeration, the coordinate forms are witnesses we consult only when the combinatorial sign rules cannot decide.
- **Condition 2 is automatic.** Planes 012 and 013 share the line $\overline{P_0P_1}$ and (since $P_3 \notin$ plane 012 by Condition 1) are distinct. They partition $\mathbb{R}^3 \setminus \overline{P_0P_1}$ into four open wedges, all non-empty. So $012^+ \cap 013^+ \neq \emptyset$ trivially. A concrete witness: any point $P_3 + \varepsilon\,\hat{n}_{013^+}$ with ε small.
- **Where it gets interesting.** Once we have all four planes $\{012, 013, 023, 123\}$ they cut $\mathbb{R}^3$ into $\binom{4}{0}+\binom{4}{1}+\binom{4}{2}+\binom{4}{3} = 15$ regions. Of the $2^4 = 16$ a-priori sign vectors, exactly one is empty — the missing 16th cell. A clean way to spot it: the sign vector that says "p is on the *opposite* side of every face from the opposite vertex of tetrahedron 0123" is not realizable (no point sees every face from the wrong side).

Algorithm: deciding if a new plane $ij(n{+}1)$ slices an existing cell

We are now ready for the inductive step. Assume we have introduced $n+1$ named points $\{P_0, \ldots, P_n\}$ and constructed the cell complex of the $\binom{n+1}{3}$ planes through every triple. We pick a new point P_{n+1} inside one specific cell C^* of the current complex; this pins the signs of all 4-brackets involving $n{+}1$ (because membership of P_{n+1} in C^* is exactly $\sigma_{C^*}(\Pi) = \mathrm{sgn}([\Pi, n{+}1])$ for every existing plane Π). Then we must slice every existing cell with the $\binom{n+1}{2}$ new planes $\{\, ij(n{+}1) : 0 \le i < j \le n \,\}$. The basic question is: given a cell C and a new plane $\Pi = ij(n{+}1)$, does Π slice C (i.e. are both $C \cap \Pi^+$ and $C \cap \Pi^-$ non-empty)?

The algorithm below answers this question using the chirotope alone (no coordinates), and is correct in the general case. The conceptual primitive is the affine identity ($\star$) from earlier — now applied to *every* 4-tuple of named points, not just $\{0,1,2,3\}$.

State. We maintain

- the **chirotope**: $\mathrm{sgn}[a,b,c,d]$ for every 4-subset $\{a,b,c,d\} \subseteq \{0,\ldots,n{+}1\}$;
- the **cell complex** of the previous step: each open cell C stored as a sign vector $\sigma_C : \mathcal{H}_{\text{old}} \to \{+,-\}$ over the old planes $\mathcal{H}_{\text{old}} = \binom{\{0,\ldots,n\}}{3}$.

We do *not* need anything else — no coordinates, no magnitudes, no "rays at infinity."

The single sign-rule. For every 4-subset $\{a,b,c,d\}$ of named indices and any point p, the affine identity ($\star$) (proved earlier for $\{0,1,2,3\}$ but valid for any 4-tuple, by the same barycentric argument) reads

$$[a,b,c,p] \;-\; [a,b,d,p] \;+\; [a,c,d,p] \;-\; [b,c,d,p] \;=\; [a,b,c,d].$$

Taking signs and noting the four magnitudes on the LHS are positive, the LHS is forced to be strictly positive (regardless of magnitudes) iff the four signs $(\sigma(abc), -\sigma(abd), \sigma(acd), -\sigma(bcd))$ are all $+$, i.e. the cell sign pattern at the four "faces of tetrahedron $abcd$" is

$$\big(\sigma(abc), \sigma(abd), \sigma(acd), \sigma(bcd)\big) \;=\; (+,-,+,-).$$

Symmetrically the LHS is forced strictly negative iff the pattern is $(-,+,-,+)$. So:

Forbidden-pattern rule (per 4-tuple). A sign vector σ violates the chirotope at $\{a,b,c,d\}$ iff

$$\big(\sigma(abc),\sigma(abd),\sigma(acd),\sigma(bcd)\big) = \big(-,+,-,+\big)\cdot \operatorname{sgn}[a,b,c,d],$$

where the right-hand side means: when $\operatorname{sgn}[a,b,c,d]=+$ the forbidden pattern is $(-,+,-,+)$, and when $\operatorname{sgn}[a,b,c,d]=-$ the forbidden pattern is $(+,-,+,-)$. This is exactly the "missing 16th cell" pattern from the original section, applied to every tetrahedron formed by four named points.

Forbidden patterns are necessary, not sufficient. A sign vector σ that corresponds to a non-empty cell of the realized arrangement necessarily avoids every forbidden pattern — this is what the affine identity $(\star_{abcd})$ proves. The converse, however, fails once $N \geq 5$: there exist sign vectors that avoid every forbidden pattern yet are not realized as cells. The first "correction-tier" is the *wedge-ordering* constraint, which we describe below.

Slice test (LP, with the forbidden-pattern rule as a pre-filter). Given an old cell C with sign vector σ_C and a new plane $\Pi = ij(n{+}1)$, we ask in turn for each candidate sign $s \in \{+1,-1\}$:

1. *Cheap chirotope check.* For every 4-subset $\{a,b,c,n{+}1\}$ such that Π is one of its four faces (i.e. $\{a,b,c\} \supseteq$ two of Π's indices), see whether the forbidden pattern fires for $\sigma_C \cup \{\Pi \mapsto s\}$ on that 4-tuple. If yes, $C \cap \Pi^s$ is provably empty — skip the LP.
2. *LP if not yet decided.* Otherwise, run an LP for feasibility of $\{\, \sigma\cdot[\cdot,p] > 0 \,\}$ where σ is σ_C extended by $\Pi \mapsto s$. If feasible, $C \cap \Pi^s$ is realized; otherwise it is not.

The plane slices C iff both halves are non-empty. The LP uses the named vertices' coordinates (one specific realization of the chirotope-so-far), not any cell witness; cell witnesses are not stored anywhere during construction.

The Farkas analysis, and where the sufficiency argument fails. The cell $C \cap \Pi^+$ is non-empty iff the strict-linear system

$$\big\{\, \sigma(abc)\,[a,b,c,p] > 0 : \text{all triples} \,\big\}$$

is feasible in $p \in \mathbb{R}^3$. By Farkas, infeasibility requires a non-negative combination $\sum \lambda_{abc}\,\sigma(abc)\,[a,b,c,p]$ that is identically ≤ 0 as a function of p. Since $[a,b,c,p]$ is affine, this means the linear part vanishes and the constant part is ≤ 0.

Differentiating $(\star_{abcd})$ in p gives one family of linear-part-zero combinations: the alternating 4-term identity

$$\mathbf{n}_{abc} \;-\; \mathbf{n}_{abd} \;+\; \mathbf{n}_{acd} \;-\; \mathbf{n}_{bcd} \;=\; 0$$

(the dependency of plane normals around the tetrahedron $\{a,b,c,d\}$). Each such dependency yields one Farkas certificate that fires exactly when its forbidden pattern is hit, so the GP rule is *necessary.*

But the tetrahedron-dependency is not the only 4- or fewer-term dependency. Consider the three planes $(i,j,a),(i,j,b),(i,j,c)$ that all contain the line $\overline{P_iP_j}$: their normals are all perpendicular to that line, hence lie in a 2-plane, hence *any three of them* are linearly dependent. This is a 3-term dependency among normals — not the alternating 4-term one — and it gives Farkas certificates that GP3 does not see. Concretely, the 3 planes through a common line cut a circular arrangement

of 6 wedges around that line; only 6 of the $2^3 = 8$ sign patterns are realized, the other 2 being forbidden. GP3 leaves all 8 admissible. This is the "wedge-ordering" constraint.

Higher-order Plücker relations contribute further constraints, and at $n \geq 7$ Mňev–Sturmfels universality applies to general realizability questions. So a complete chirotope-only characterization of topes is real and hard.

What this means for the algorithm. GP3 is a fast pre-filter: when it fires, the extension is provably empty and we can skip the LP. When it does not fire, we have to ask the LP (using the vertex realizations as input data) for the final yes/no. The LP is exact and chirotope-respecting — different realizations of the same chirotope give the same yes/no answer — so the LP-based slicer enumerates the correct cell complex. The vertex witnesses are needed by the LP only as a specific representative of the chirotope; the cell complex shape is intrinsic to the chirotope.

Why this is still not NP-complete in our setting. Realizability of an *arbitrary* abstract chirotope is $\exists\mathbb{R}$-complete (Mnëv), but that is not the question we ask: by construction we always extend a chirotope that is already realized. For an already-realized configuration, each slice test is just LP feasibility, which lies far below the worst-case complexity wall.

Per-step complexity. For step $n \to n{+}1$: $\binom{n+1}{2}$ new planes, each tested against every old cell. The number of old cells is bounded by Zaslavsky's count, $O(n^9)$. Each slice test is a single LP feasibility check (or zero, when the GP forbidden-pattern pre-filter fires), which is polynomial-time. The whole step is polynomial in n, dominated by the cell count.

Segment-triangle piercing in bracket form

A natural geometric primitive (and a check we will need later, e.g. for "does this Császár candidate actually embed without crossings?"): given two segments and a triangle made of named points, does the segment $\overline{P_iP_j}$ pierce the (open) triangle $\triangle P_kP_lP_m$? The answer is a four-line test in the chirotope, no coordinates needed.

Two conditions. Assume the five points are in general position (no 4 coplanar, so all five 4-brackets on $\{i, j, k, l, m\}$ are nonzero). The segment pierces the triangle iff both:

1. *The line through P_i, P_j pierces the triangle's interior*, i.e. the three signed volumes of segment-against-edge tetrahedra all agree:

$$\operatorname{sgn}[i, j, k, l] \;=\; \operatorname{sgn}[i, j, l, m] \;=\; \operatorname{sgn}[i, j, m, k] \;\in\; \{+, -\}.$$

   Each $[i, j, \cdot, \cdot]$ is, up to sign, the signed volume of the tetrahedron formed by the segment and one edge of the triangle. Their common sign records the directed orientation of $\overline{P_iP_j}$ "going around" the triangle. If any one differs, the line misses or grazes that edge.

2. *The two endpoints P_i, P_j lie on opposite sides of the plane through klm*, so the crossing point of the line with the plane falls inside the segment:

$$\operatorname{sgn}[k, l, m, i] \;\neq\; \operatorname{sgn}[k, l, m, j].$$

So the piercing test reads off five 4-bracket signs on $\{i, j, k, l, m\}$ and checks two equalities. The five brackets are tied by the GP identity $\sum_q (-1)^q [\widehat{q}] = 0$, so any one is determined up to sign by the other four; in practice we just take whichever signs the chirotope already has.

Two-segment intersection. As an aside: two segments $\overline{P_iP_j}$ and $\overline{P_kP_l}$ in $\mathbb{R}^3$ can meet inside their interiors only if they are coplanar, which requires $[i,j,k,l]=0$ — a non-generic condition. So in the generic chirotope two named segments share at most a common endpoint, and $\mathrm{sgn}[i,j,k,l]$ records which of the two segments "goes above" the other (the sign of the lift).

Redundant halfspaces in cell descriptions

A cell of our arrangement is described by signs on *every* plane, but only some of these signs are *essential*: dropping a non-essential one leaves the cell unchanged. A non-essential constraint is *redundant*; the planes whose constraints are essential are the *facets* of the cell.

The tetrahedron example. In the four-plane arrangement of tetrahedron 0123, the cell "past P_3" (the open cone past vertex P_3, bounded below by the three faces meeting at P_3) has sign vector

$$\big(\sigma(012),\sigma(013),\sigma(023),\sigma(123)\big)=(+,+,-,+).$$

It really has only three facets — the three faces $013, 023, 123$ that meet at P_3. The fourth sign $\sigma(012)=+$ is redundant: the cell happens to lie on the same side of 012 as the tetrahedron's interior, but the plane 012 is not a face of the cell. Equivalently, flipping $\sigma(012)$ to $-$ would give the sign vector $(-,+,-,+)$, which is precisely the missing 16th cell — empty. So this constraint cannot be the binding one.

The chirotope-only redundancy test. The previous paragraph generalizes: *the constraint σ_j on plane $\Pi_j = abc$ is redundant for a cell C iff flipping it produces an empty cell, iff some forbidden-pattern rule is triggered after the flip.* Concretely,

> $\sigma(abc)$ is redundant for C iff there is some $d\notin\{a,b,c\}$ such that, after flipping $\sigma(abc)$, the four signs $\big(\sigma(abc),\sigma(abd),\sigma(acd),\sigma(bcd)\big)$ form the forbidden pattern $(-,+,-,+)\cdot\mathrm{sgn}[a,b,c,d]$ of the 4-tuple $\{a,b,c,d\}$.

This is a pure chirotope check, with $O(n)$ work per plane (one 4-tuple per choice of d). It tells us, for each cell, exactly which sign-constraints are facets and which are mere consequences of the rest.

Speeding up cell splitting. The same machinery serves as a pre-filter for the slice test. To decide whether a new plane Π slices cell C, we previously had to ask the LP about both extensions σ^+,σ^-. With the chirotope filter:

1. Run the forbidden-pattern check on σ^+ and on σ^- in $O(n)$ time, considering only the 4-tuples that involve Π's three indices.
2. If both σ^+ and σ^- are forbidden-free: Π slices C, do the actual split.
3. If exactly one is forbidden: the cell lies entirely on the other side; attach that sign to C and move on without an LP.
4. Both being forbidden cannot happen for an existing cell — C has a witness which already realizes one extension.

In our $n=7$ run with ~ 3000 cells, the new-plane stage at $n=8$ would ask $\sim 3000\times 21$ slice questions; the filter answers most of them without any LP, since for most cells most new planes are redundant.

Storing only facets. As a representational optimization, each cell can be stored as the dict of its *facet* signs (with the rest re-derived from the chirotope on demand). With this convention, the interior of tetrahedron 0123 lists all four signs (four facets); the past-P_3 cone lists only three; and the forbidden 16th cell, if it were realized, would list zero — which is precisely the way to recognize that it is empty.